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Course: Networks System Design

Instructor: KUY Movsun

Assignment: Lab10 (The Link Layer and LANs)

Due Date: January, 20 2026, (12:00 AM)

Part 1: Manual Calculation

Exercise 1: Cyclic Redundancy Check (CRC)

Concept

CRC is used by network hardware to detect bit errors in transmitted frames. The sender computes CRC bits **R** such that the combined bit pattern (D, R) is exactly divisible by the generator polynomial **G** using modulo-2 arithmetic (XOR).

Given

- **Data Packet (D)** = 101010 (6 bits)
 - **Generator Polynomial (G)** = 1001 (4 bits)
 - **CRC length (r)** = $\text{length}(G) - 1 = 3 \text{ bits}$
-

Step 1: Append Zeros to Data

Append 3 zeros to the data packet: 101010000

Step 2: Modulo-2 Division

Perform binary (XOR) division of 101010000 by 1001.

Resulting remainder: R = 111

Step 3: Transmitted Bit Pattern

Append the remainder to the original data: <D, R> = 101010111

Step 4: Verification

Divide the transmitted pattern **101010111** by the generator **1001**. Remainder = 000

✓ The remainder is zero, so the frame is valid and error-free.

Step 5: Simulate an Error

Flip the last bit of the transmitted pattern:

- Original frame: 101010111
- Error frame: 101010110

Divide the erroneous frame by **1001**: Remainder $\neq$ 000

Analysis (CRC)

- The remainder is **non-zero**, meaning an error is detected.
 - The receiver uses this result to **discard the frame** and rely on higher-layer protocols for retransmission.
 - This is how CRC enables reliable error detection in hardware.
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Exercise 2: CSMA/CD Exponential Backoff

Concept

Ethernet uses CSMA/CD with binary exponential backoff to reduce repeated collisions. After the m -th collision, a node chooses a random value:

$$[K \in \{0, 1, 2, \dots, 2^m - 1\}]$$

The node waits **$K \times 512$ bit times** before retransmitting.

Given

- Collision count: **$m = 3$**
 - Chosen random value: **$K = 4$**
 - Ethernet speed: **10 Mbps**
 - 1 bit time = **$0.1 \mu\text{s}$**
-

Step 1: Collision Window

$$[2^3 - 1 = 7]$$

Possible values of **K** : {0, 1, 2, 3, 4, 5, 6, 7}

Step 2: Wait Time in Bit Times

$$[4 \times 512 = 2048 \text{ \textit{bit times}}]$$

Step 3: Convert to Real Time

$$[2048 \times 0.1 \mu s = 204.8 \mu s]$$

Total wait time before retransmission: 204.8 microseconds

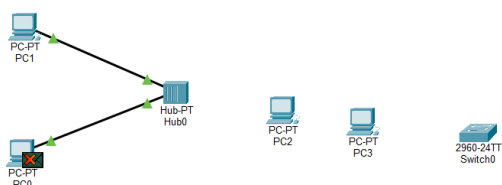
Analysis (CSMA/CD)

- As **m increases**, the random backoff range becomes larger.
- This spreads retransmission attempts over time.
- It reduces repeated collisions and helps **relieve network congestion** when traffic is heavy.

Activity A: Shared Media and Collisions with Hubs (Part 2)

Topology Setup

- **Devices Used:**
 - PC0, PC1, PC2, PC3
 - Hub0
 - Switch0
- **Connections:**
 - PC0 and PC1 connected to Hub0 using Copper Straight-Through cables
 - PC2 and PC3 not connected
- **Simulation Mode:**
 - Switch to Simulation Mode (Shift + S)
 - Open "Edit Filters" and enable only **ICMP**
- **Traffic Generation:**
 - Use "Add Simple PDU" tool: - PC0 → PC1 - PC1 → PC0



Event List				
Vis.	Time(sec)	Last Device	At Device	Type
...	0.000	--	PC1	ICMP
...	0.000	--	PC0	ICMP
...	0.002	Hub0	PC0	ARP
...	0.002	Hub0	PC1	ARP
...	2.001	--	PC1	ICMP
...	2.003	--	PC0	ICMP

Reset Simulation ☒ Constant Delay Captures 1013.11

Play Controls

- **Observation:**
 - Step through simulation using "Play" or "Forward"
 - Watch for collision icon at Hub0 (fire or red X)

Event List Summary

Time (sec)	Last Device	At Device	Packet Type
0.000	—	PC1	ICMP
0.000	—	PC0	ICMP
0.002	Hub0	PC0	ARP
0.002	Hub0	PC1	ARP
2.001	—	PC1	ICMP
2.003	—	PC0	ICMP

Analysis Question

Why does a collision occur when using a Hub?

- **Layer 1 Behavior (Hub):**
 - Operates at the Physical Layer
 - Does not inspect MAC addresses
 - Repeats incoming signals to all ports
 - Creates a **shared collision domain**
 - When two devices transmit simultaneously, signals overlap → **collision**
- **Layer 2 Behavior (Switch):**
 - Operates at the Data Link Layer
 - Learns MAC addresses and forwards frames only to the correct port
 - Each port is its own collision domain
 - **No collisions** occur in full-duplex switch environments

Conclusion

Hubs cause collisions due to their broadcast nature at Layer 1. Switches prevent collisions by intelligently forwarding frames at Layer 2.

Activity B: Switch MAC Table Learning

Objective

Understand how switches learn device locations to create **dedicated collision domains**, unlike hubs which broadcast to all devices.

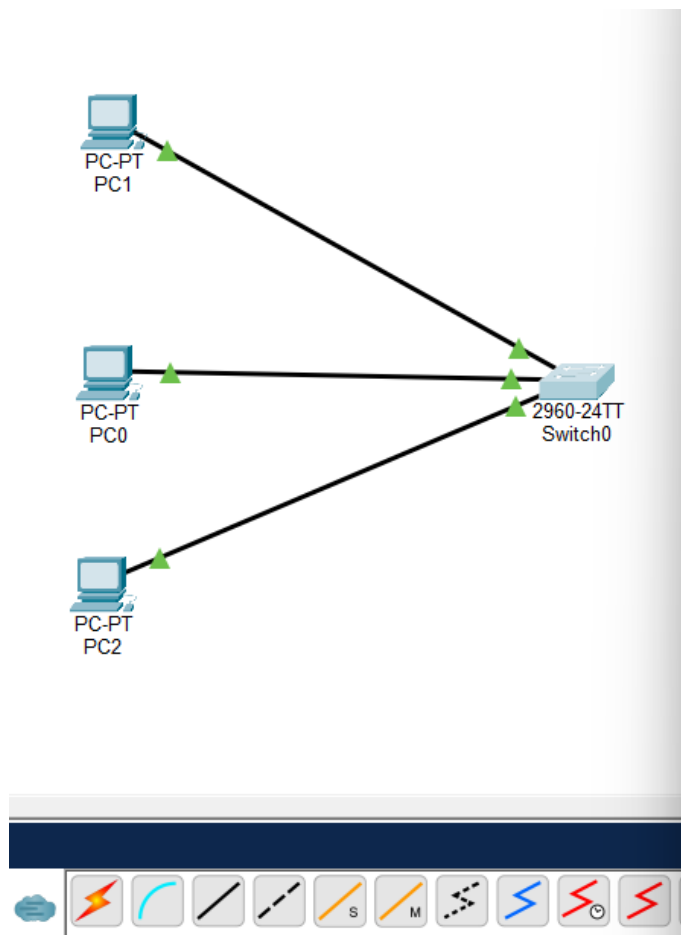
Steps

1. Re-Connect

- Delete the Hub from the previous activity.
- Connect **PC0, PC1, and PC2** to the Switch.
- Wait for link lights to turn green (or press **Fast Forward Time** in Realtime mode).

2. Inspect Empty Table

- Open the Switch CLI.
- Run: show mac-address-table
- The table should be **empty** initially.



```

Top Assembly Revision Number   : A0
Version ID                     : V02
CLEI Code Number              : COM3L00BRA
Hardware Board Revision Number : 0x01

Switch Ports Model          SW Version
-----
*      1 26      WS-C2960-24TT-L  15.0(2)SE4

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M)
Technical Support: http://www.cisco.com/techsupport
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Compiled Wed 26-Jun-13 02:49 by mnguyen

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/1, changed s
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastE
%LINK-5-CHANGED: Interface FastEthernet0/2, changed s
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastE
%LINK-5-CHANGED: Interface FastEthernet0/3, changed s
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastE

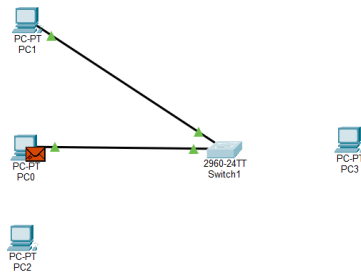
Switch>show mac-address-table
                Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----
Switch>
  
```

3. Ping

- Switch to **Simulation Mode**.
- Edit Filters: Ensure **ARP** and **ICMP** are checked.
- Use the **Add Simple PDU** tool to ping from **PC0** → **PC1**.

4. Step-by-Step Learning

- Click **Forward** until the frame leaves PC0 and reaches the Switch.
 - Stop and check the MAC table again: show mac-address-table
 - PC0's MAC address should now be associated with its port (e.g., **Fa0/1**).
- The switch has **learned PC0's location**.



Event List				
Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	ICMP
	0.000	--	PC0	ARP
	0.001	PC0	Switch1	ARP
	2.003	--	PC0	ICMP

Reset Simulation ☒ Constant Delay

Play Controls

Captured to: 1760.279 s

5. Complete the Ping

- Continue forwarding until the reply returns from PC1 to PC0.
- Check the MAC table again:
- PC1's MAC address should now be learned and associated with its port.

Analysis Question

In Activity A (Hub), the packet was broadcast to everyone. In this activity, after the MAC table was populated, did the Switch send the ICMP reply (from PC1) to PC2? Why or why not?

- **Answer:** No, the Switch did **not** send the ICMP reply to PC2.
- **Reason:** Once the MAC table is populated, the Switch knows which port corresponds to PC0's MAC address. It forwards the reply **only to PC0's port**.
- **Key Difference:** Unlike a Hub, which broadcasts frames to all devices, a Switch isolates traffic per port using its MAC table. This prevents collisions and unnecessary broadcasts.

Conclusion

Switches dynamically build a MAC address table by learning source addresses from incoming frames. This allows them to forward traffic only to the correct destination port, creating **collision-free communication** and efficient use of bandwidth.

Activity C: Switch MAC Table Learning

Objective

Understand how switches learn MAC addresses to create dedicated collision domains and forward frames intelligently.

Setup Instructions

1. Reconnect Devices

- Delete the Hub from previous activities.
- Connect **PC0, PC1, and PC2** to **Switch0** using Copper Straight-Through cables.
- Wait for link lights to turn green or press **Fast Forward Time** in Realtime mode.

2. Inspect the Empty MAC Table

- Click on the Switch.
- Go to the **CLI tab**.
- Type: show mac-address-table
- The table should be **empty** initially.

3. Generate Traffic

- Switch to **Simulation Mode**.
- Open **Edit Filters** and ensure **ARP** and **ICMP** are checked.
- Use the **Add Simple PDU** tool to ping from **PC0** → **PC1**.

Step-by-Step MAC Learning

4. Observe Frame Movement

- Click **Forward** until the frame leaves PC0 and reaches the Switch.

5. Check MAC Table

- Open the Switch CLI again and run: show mac-address-table
 - You should now see **PC0's MAC address** associated with its port (e.g., Fa0/1).

6. Complete the Ping

- Continue clicking **Forward** until the ICMP reply returns from PC1 to PC0.

7. Final MAC Table Check

- Run: show mac-address-table
- You should now see **PC1's MAC address** added to the table.

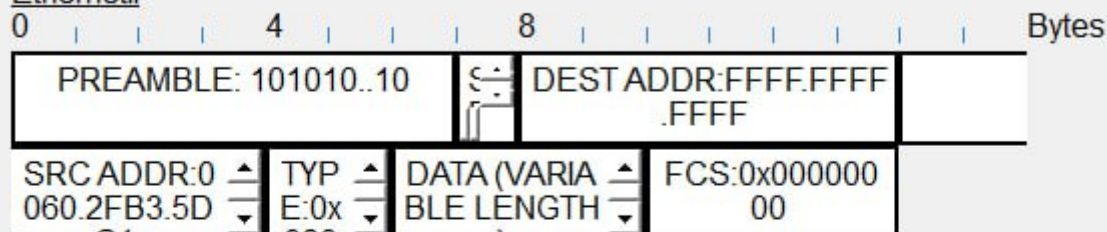
```
Switch>show mac-address-table
Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----
Switch>show mac-address-table
Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----
1       0060.2fb3.5dc1   DYNAMIC   Fa0/1
1       00e0.b053.725c   DYNAMIC   Fa0/2
```

PDU Information at Device: PC0

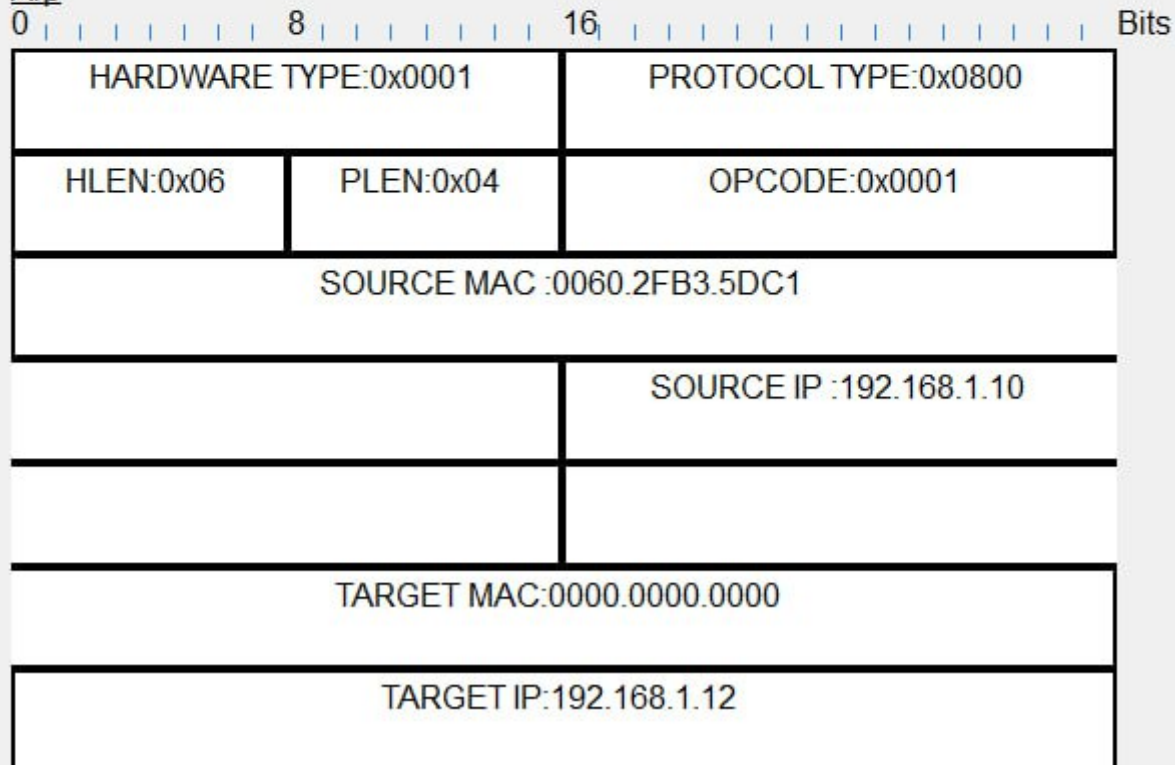
OSI Model [Outbound PDU Details](#)

PDU Formats

EthernetII



Arp



Event List				
Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	PC0	ARP
	0.000	--	PC0	ICMP
	0.000	--	PC0	ARP
	0.001	PC0	Switch0	ARP
	0.001	--	PC0	ARP
	0.002	PC0	Switch0	ARP
	0.002	Switch0	PC1	ARP
	0.002	Switch0	PC2	ARP
	0.003	Switch0	PC1	ARP
	0.003	Switch0	PC2	ARP
	0.004	PC2	Switch0	ARP
	0.005	Switch0	PC0	ARP
	0.005	--	PC0	ICMP
	0.006	PC0	Switch0	ICMP
	0.007	Switch0	PC2	ICMP
	0.008	PC2	Switch0	ICMP
	0.009	Switch0	PC0	ICMP

Analysis Question

In Activity A (Hub), the packet was broadcast to everyone. In this activity, after the MAC table was populated, did the Switch send the ICMP reply (from PC1) to PC2? Why or why not?

- **Answer:** No, the Switch did **not** send the ICMP reply to PC2.
- **Reason:** The Switch had already learned PC0's MAC address and its associated port. It forwarded the reply **only to PC0**, not to PC2.
- **Conclusion:** Unlike a Hub, a Switch uses its MAC table to forward frames only to the correct destination, preventing unnecessary broadcasts and collisions.

Summary

Switches dynamically learn MAC addresses from incoming frames and build a table to forward traffic efficiently. This isolates communication between devices and eliminates collisions, making switches ideal for modern LAN environments.